



Syllabus: Random variables, Uniform, Normal, Exponential, Poisson and Binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	1	0	2	1	0	0	1	1	0
2 Marks Count	1	1	1	2	1	1	1	0	2	2	0
Total Marks	3	3	3	4	4	3	2	0	5	5	0

Welcome to the "Engineering Mathematics: Probability" chapter, a cornerstone for any aspiring GATE Computer Science candidate. This subject delves into the mathematical framework for quantifying uncertainty, analyzing random phenomena, and making informed decisions in the face of incomplete information. For GATE CS, a solid understanding of probability is indispensable, as it forms the theoretical bedrock for crucial areas like algorithms (randomized algorithms, average-case analysis), data structures (hash tables, skip lists), machine learning (classification, regression, neural networks, Bayesian inference), artificial intelligence, networking (queueing theory, packet loss), and even operating systems (scheduling, resource allocation). Typically, probability questions constitute a significant portion of the Engineering Mathematics section, often contributing 3-5 marks directly, with concepts frequently integrated into 1-mark or 2-mark questions across other CS topics. Questions can range from direct formula application and conceptual understanding to problem-solving involving combinations of distributions and theorems, often requiring careful interpretation of scenarios.

Topic-wise Key Concepts

Bayes Theorem

Definition and Core Idea: Bayes' Theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event. It's a fundamental concept for updating beliefs or probabilities as new evidence becomes available, forming the basis of Bayesian inference.

• Important Formulas, Theorems, and Results:

1. **Bayes' Theorem:** For two events A and B , where $P(B) > 0$,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where $P(B)$ can be expanded using the law of total probability:

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

More generally, for a partition $A_1, A_2, \dots, A_n$ of the sample space,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^n P(B|A_j)P(A_j)}$$

• Key Properties and Identities:

- Allows for the calculation of posterior probability $P(A|B)$ from prior probability $P(A)$ and likelihood $P(B|A)$.
- Crucial for diagnostic testing, spam filtering, and machine learning algorithms (e.g., Naive Bayes).

• Common Pitfalls or Tricky Points:

- Confusing $P(A|B)$ with $P(B|A)$.
- Incorrectly calculating the denominator $P(B)$ by not considering all mutually exclusive and exhaustive cases.
- Not identifying the correct prior probabilities $P(A_i)$ and conditional probabilities $P(B|A_i)$.

• Standard Problem-Solving Techniques or Shortcuts:

- Clearly define events A and B .
- Break down the problem into identifying $P(A)$, $P(B|A)$, and $P(B|A^c)$ (or $P(B|A_i)$).
- Use tree diagrams to visualize conditional probabilities and total probability.

Bayesian Network

Definition and Core Idea: A Bayesian Network (BN) is a probabilistic graphical model that represents a set of random

variables and their conditional dependencies via a directed acyclic graph (DAG). Nodes represent random variables, and directed edges represent conditional dependencies, allowing for efficient representation and inference of joint probability distributions.

- **Important Formulas, Theorems, and Results:**

1. **Joint Probability Distribution:** For a Bayesian Network with nodes $X_1, \dots, X_n$, the joint probability distribution is given by the product of the conditional probability distributions of each node given its parents:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$$

2. **Conditional Independence:** A node is conditionally independent of its non-descendants given its parents.

- **Key Properties and Identities:**

- Provides a compact representation of joint probability distributions, especially for many variables.
- Facilitates inference (e.g., calculating $P(X|E)$ where E is evidence).
- The graph structure explicitly shows the assumed conditional independencies.

- **Common Pitfalls or Tricky Points:**

- Misinterpreting the direction of edges or the meaning of conditional independence.
- Incorrectly applying the chain rule for joint probability without respecting the network's structure.
- Confusing marginal independence with conditional independence.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Identify parents for each node to write down the joint probability.
- Use d-separation rules to determine conditional independencies graphically.
- For inference, apply variable elimination or message passing algorithms (though GATE questions are usually simpler, focusing on structure and basic inference).

Bernoulli Distribution

Definition and Core Idea: The Bernoulli distribution is a discrete probability distribution for a random variable that takes value 1 with probability p (success) and value 0 with probability $1 - p$ (failure). It models a single trial of an experiment with only two possible outcomes.

- **Important Formulas, Theorems, and Results:**

1. **Probability Mass Function (PMF):** For a Bernoulli random variable X ,

$$P(X = x) = p^x(1 - p)^{1-x} \quad \text{for } x \in \{0, 1\}$$

2. **Mean (Expectation):** $E[X] = p$

3. **Variance:** $\text{Var}(X) = p(1 - p)$

- **Key Properties and Identities:**

- It's the simplest discrete distribution.
- The sum of n independent Bernoulli trials follows a Binomial distribution.

- **Common Pitfalls or Tricky Points:**

- Confusing Bernoulli with Binomial (Bernoulli is a single trial, Binomial is multiple trials).
- Incorrectly identifying p (probability of success).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Identify the "success" event and its probability p .
- Directly apply formulas for mean and variance.

Binomial Distribution

Definition and Core Idea: The Binomial distribution models the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success p . It is a discrete probability distribution.

- **Important Formulas, Theorems, and Results:**

1. **Probability Mass Function (PMF):** For $X \sim B(n, p)$,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k \in \{0, 1, \dots, n\}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient.

2. **Mean (Expectation):** $E[X] = np$
3. **Variance:** $\text{Var}(X) = np(1 - p)$
4. **Standard Deviation:** $\sigma = \sqrt{np(1 - p)}$
- **Key Properties and Identities:**
 - Sum of n independent Bernoulli random variables.
 - Symmetric if $p = 0.5$, skewed otherwise.
 - Approximated by Poisson distribution when n is large and p is small (and np is moderate).
 - Approximated by Normal distribution when n is large (and $np > 5, n(1 - p) > 5$).
- **Common Pitfalls or Tricky Points:**
 - Incorrectly identifying n (number of trials) or p (probability of success).
 - Miscalculating binomial coefficients.
 - Confusing "at least," "at most," "exactly" in problem statements (e.g., $P(X \geq k)$ vs. $P(X = k)$).
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Identify if the problem involves a fixed number of independent trials with two outcomes.
 - Use the complement rule for "at least" or "at most" probabilities (e.g., $P(X \geq k) = 1 - P(X < k)$).
 - Memorize mean and variance formulas.

Chi Square Distribution

Definition and Core Idea: The Chi-Square (χ^2) distribution is a continuous probability distribution that arises in statistics, particularly in hypothesis testing and confidence interval estimation. It is the distribution of a sum of the squares of independent standard normal random variables.

- **Important Formulas, Theorems, and Results:**
 1. **Definition:** If $Z_1, Z_2, \dots, Z_k$ are independent standard normal random variables (i.e., $Z_i \sim N(0, 1)$), then the random variable $X = \sum_{i=1}^k Z_i^2$ follows a Chi-Square distribution with k degrees of freedom, denoted $X \sim \chi^2(k)$.
 2. **Probability Density Function (PDF):**

$$f(x; k) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{(k/2)-1} e^{-x/2} \quad \text{for } x > 0$$

where Γ is the Gamma function.

3. **Mean (Expectation):** $E[X] = k$
4. **Variance:** $\text{Var}(X) = 2k$
- **Key Properties and Identities:**
 - Skewed to the right, but becomes more symmetric as k increases.
 - Used in goodness-of-fit tests, tests of independence, and for variance estimation.
 - The sum of independent chi-square random variables is also a chi-square random variable with degrees of freedom equal to the sum of their individual degrees of freedom.
- **Common Pitfalls or Tricky Points:**
 - Incorrectly identifying the degrees of freedom k .
 - Confusing the Z_i^2 definition with other squared terms.
 - GATE questions usually focus on its properties or its role in statistical tests rather than direct PDF calculations.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Understand that it's a sum of squared standard normal variables.
 - Memorize mean and variance in terms of degrees of freedom.
 - Recognize its application in hypothesis testing contexts (e.g., for categorical data).

Conditional Probability

Definition and Core Idea: Conditional probability is the probability of an event occurring given that another event has already occurred. It quantifies how the probability of an event changes when new information is available.

- **Important Formulas, Theorems, and Results:**
 1. **Definition:** For two events A and B , with $P(B) > 0$, the conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2. Multiplication Rule of Probability:

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

3. **Law of Total Probability:** If $A_1, A_2, \dots, A_n$ form a partition of the sample space, then for any event B :

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

- **Key Properties and Identities:**

- If A and B are independent, $P(A|B) = P(A)$.
- $P(A|A) = 1$.
- $P(A^c|B) = 1 - P(A|B)$.

- **Common Pitfalls or Tricky Points:**

- Incorrectly identifying $P(A \cap B)$ or $P(B)$.
- Confusing $P(A|B)$ with $P(B|A)$ or $P(A \cap B)$.
- Not recognizing when events are independent, which simplifies calculations.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Clearly define the events A and B .
- Use Venn diagrams or tree diagrams to visualize the sample space and events.
- Break down complex problems into simpler conditional probabilities.

Continuous Distribution

Definition and Core Idea: A continuous probability distribution describes the probabilities of a random variable taking on values in a continuous range (e.g., real numbers). Unlike discrete distributions, the probability of a continuous random variable taking on any single exact value is zero.

- **Important Formulas, Theorems, and Results:**

1. **Probability Density Function (PDF):** $f(x)$ such that $f(x) \geq 0$ for all x , and $\int_{-\infty}^{\infty} f(x)dx = 1$.
2. **Probability of an Interval:** $P(a \leq X \leq b) = \int_a^b f(x)dx$.
3. **Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$.
4. **Relationship between PDF and CDF:** $f(x) = \frac{d}{dx}F(x)$ (where $F(x)$ is differentiable).
5. **Mean (Expectation):** $E[X] = \int_{-\infty}^{\infty} xf(x)dx$.
6. **Variance:** $\text{Var}(X) = E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} (x - E[X])^2 f(x)dx$.

- **Key Properties and Identities:**

- $P(X = x) = 0$ for any specific value x .
- The area under the PDF curve over its entire range is 1.
- The CDF is non-decreasing and ranges from 0 to 1.

- **Common Pitfalls or Tricky Points:**

- Confusing PDF with PMF (for discrete distributions).
- Incorrectly calculating integrals for probabilities, mean, or variance.
- Forgetting that $P(a \leq X \leq b) = P(a < X < b)$ for continuous distributions.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Always check if the given function is a valid PDF (non-negative and integrates to 1).
- Use integration for calculating probabilities and moments.
- Understand the relationship between PDF and CDF.

Expectation

Definition and Core Idea: The expectation (or expected value, mean) of a random variable is the weighted average of all possible values that the random variable can take, with the weights being their respective probabilities. It represents the "long-run average" or central tendency of the random variable.

- **Important Formulas, Theorems, and Results:**

1. **For Discrete Random Variable X :**

$$E[X] = \sum_x xP(X = x)$$

2. For Continuous Random Variable X :

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

3. Expectation of a Function of a Random Variable $g(X)$:

- Discrete: $E[g(X)] = \sum_x g(x)P(X = x)$
- Continuous: $E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$

4. Linearity of Expectation: For any random variables X, Y and constants a, b :

$$E[aX + bY] = aE[X] + bE[Y]$$

5. Expectation of Product (for Independent Variables): If X and Y are independent, then

$$E[XY] = E[X]E[Y].$$

• Key Properties and Identities:

- $E[c] = c$ for a constant c .
- $E[X - E[X]] = 0$.
- Expectation is a linear operator.

• Common Pitfalls or Tricky Points:

- Confusing $E[X^2]$ with $(E[X])^2$. They are generally not equal.
- Applying $E[XY] = E[X]E[Y]$ when X and Y are not independent.
- Incorrectly summing or integrating over the wrong range.

• Standard Problem-Solving Techniques or Shortcuts:

- Identify whether the variable is discrete or continuous.
- Use linearity of expectation extensively, as it holds even for dependent variables.
- For complex functions, break them down using linearity.

Exponential Distribution

Definition and Core Idea: The Exponential distribution is a continuous probability distribution that describes the time between events in a Poisson process, i.e., a process in which events occur continuously and independently at a constant average rate. It is memoryless, meaning the past duration does not affect future duration.

• Important Formulas, Theorems, and Results:

1. Probability Density Function (PDF): For $X \sim \text{Exp}(\lambda)$,

$$f(x; \lambda) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

where $\lambda > 0$ is the rate parameter.

2. Cumulative Distribution Function (CDF):

$$F(x; \lambda) = P(X \leq x) = 1 - e^{-\lambda x} \quad \text{for } x \geq 0$$

3. Mean (Expectation): $E[X] = \frac{1}{\lambda}$

4. Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$

5. Memoryless Property: $P(X > s + t | X > s) = P(X > t)$ for any $s, t \geq 0$.

• Key Properties and Identities:

- Models waiting times, lifetimes of components, etc.
- Its rate parameter λ is the inverse of the mean.
- The only continuous distribution with the memoryless property.

• Common Pitfalls or Tricky Points:

- Confusing the rate parameter λ with the mean $1/\lambda$.
- Incorrectly applying the memoryless property or forgetting its implications.
- Errors in integration for probabilities or moments if not using CDF.

• Standard Problem-Solving Techniques or Shortcuts:

- Identify problems involving "time until" an event or "lifetime" with a constant rate.
- Directly use CDF for $P(X \leq x)$ or $P(X > x)$.
- Apply the memoryless property to simplify conditional probability questions.

Independent Events

Definition and Core Idea: Two events are independent if the occurrence of one does not affect the probability of the other occurring. Their probabilities multiply to give the probability of both occurring.

• **Important Formulas, Theorems, and Results:**

1. **Definition:** Events A and B are independent if and only if:

$$P(A \cap B) = P(A)P(B)$$

2. **Equivalent Conditions:** If $P(B) > 0$, then A and B are independent if and only if $P(A|B) = P(A)$. Similarly, if $P(A) > 0$, then $P(B|A) = P(B)$.
3. **Independence of Complements:** If A and B are independent, then A and B^c , A^c and B , and A^c and B^c are also independent.

• **Key Properties and Identities:**

- Simplifies calculations involving joint probabilities.
- Crucial assumption for many statistical models and distributions (e.g., Binomial, Poisson).

• **Common Pitfalls or Tricky Points:**

- Confusing independent events with mutually exclusive events (disjoint events). Mutually exclusive events cannot be independent unless one has zero probability.
- Assuming independence when it's not explicitly stated or logically implied.
- Incorrectly applying the product rule for dependent events.

• **Standard Problem-Solving Techniques or Shortcuts:**

- Always check the definition $P(A \cap B) = P(A)P(B)$ to verify independence.
- If events are independent, use the simplified conditional probability $P(A|B) = P(A)$.

Normal Distribution

Definition and Core Idea: The Normal (or Gaussian) distribution is a symmetric, bell-shaped continuous probability distribution that is ubiquitous in natural and social sciences. It is characterized by its mean (μ) and standard deviation (σ), and is central to the Central Limit Theorem.

• **Important Formulas, Theorems, and Results:**

1. **Probability Density Function (PDF):** For $X \sim N(\mu, \sigma^2)$,

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

2. **Mean (Expectation):** $E[X] = \mu$
3. **Variance:** $\text{Var}(X) = \sigma^2$
4. **Standard Deviation:** $\text{SD}(X) = \sigma$
5. **Standard Normal Distribution:** A normal distribution with $\mu = 0$ and $\sigma = 1$, denoted $Z \sim N(0, 1)$.
6. **Standardization:** If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.
7. **Empirical Rule (68-95-99.7 Rule):** Approximately 68% of data falls within 1σ of the mean, 95% within 2σ , and 99.7% within 3σ .

• **Key Properties and Identities:**

- Symmetric about its mean.
- The mean, median, and mode are all equal to μ .
- Sum of independent normal random variables is also normal.
- Central Limit Theorem: The sum/average of a large number of independent and identically distributed random variables, regardless of their original distribution, tends towards a normal distribution.

• **Common Pitfalls or Tricky Points:**

- Not standardizing correctly before using Z-tables (if tables were provided, though GATE usually tests concepts).
- Confusing variance σ^2 with standard deviation σ .
- Assuming normality when not justified, especially for small sample sizes.

• **Standard Problem-Solving Techniques or Shortcuts:**

- Convert any normal variable to a standard normal variable using $Z = (X - \mu)/\sigma$.
- Use the symmetry of the normal distribution: $P(Z < -z) = P(Z > z)$.
- Apply the empirical rule for quick estimations.

Poisson Distribution

Definition and Core Idea: The Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.

• **Important Formulas, Theorems, and Results:**

1. **Probability Mass Function (PMF):** For $X \sim \text{Pois}(\lambda)$,

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k \in \{0, 1, 2, \dots\}$$

where $\lambda > 0$ is the average rate of events in the interval.

2. **Mean (Expectation):** $E[X] = \lambda$

3. **Variance:** $\text{Var}(X) = \lambda$

• **Key Properties and Identities:**

- Models rare events.
- The mean and variance are equal (λ).
- Can approximate the Binomial distribution when n is large and p is small, with $\lambda = np$.
- The sum of independent Poisson random variables is also a Poisson random variable (with rate parameter equal to the sum of individual rates).

• **Common Pitfalls or Tricky Points:**

- Incorrectly identifying λ (the average rate for the specified interval). Ensure λ is consistent with the time/space unit of the problem.
- Miscalculating factorials or powers of λ .
- Confusing "at least," "at most," "exactly" in problem statements.

• **Standard Problem-Solving Techniques or Shortcuts:**

- Identify problems involving counts of events over a fixed interval.
- Use the complement rule for "at least" probabilities.
- Remember that mean and variance are both λ .

Probability

Definition and Core Idea: Probability is a numerical measure of the likelihood of an event occurring. It is a value between 0 and 1, where 0 indicates impossibility and 1 indicates certainty. It forms the foundation for understanding randomness and uncertainty.

• **Important Formulas, Theorems, and Results:**

1. **Axioms of Probability:**

- For any event A , $0 \leq P(A) \leq 1$.
- $P(S) = 1$ where S is the sample space.
- For a sequence of mutually exclusive events $A_1, A_2, \dots$, $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

2. **Complement Rule:** $P(A^c) = 1 - P(A)$.

3. **Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

4. **For Mutually Exclusive Events:** If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

• **Key Properties and Identities:**

- Probabilities are non-negative.
- The sum of probabilities of all possible outcomes is 1.
- The probability of the impossible event is 0.

• **Common Pitfalls or Tricky Points:**

- Incorrectly identifying the sample space or events.
- Misapplying the addition rule without accounting for overlaps.
- Confusing "and" ($\cap$) with "or" ($\cup$).

• **Standard Problem-Solving Techniques or Shortcuts:**

- Clearly define the sample space and events.
- Use Venn diagrams for visualizing events and their intersections/unions.
- For equally likely outcomes, $P(A) = \frac{\text{Number of outcomes in } A}{\text{Total number of outcomes}}$.

Probability Density Function (PDF)

Definition and Core Idea: The Probability Density Function (PDF), denoted $f(x)$, is a function associated with continuous random variables. It describes the relative likelihood for the random variable to take on a given value. The area under the PDF curve over an interval gives the probability that the variable falls within that interval.

- **Important Formulas, Theorems, and Results:**

1. **Properties of a PDF:**

- $f(x) \geq 0$ for all x .
- $\int_{-\infty}^{\infty} f(x)dx = 1$.

2. **Probability Calculation:** $P(a \leq X \leq b) = \int_a^b f(x)dx$.

3. **Relationship with CDF:** $f(x) = \frac{d}{dx}F(x)$, where $F(x)$ is the Cumulative Distribution Function.

- **Key Properties and Identities:**

- The PDF itself does not give a probability; rather, its integral over an interval does.
- For continuous variables, $P(X = x) = 0$.

- **Common Pitfalls or Tricky Points:**

- Interpreting $f(x)$ as $P(X = x)$. This is incorrect for continuous variables.
- Forgetting to check if the PDF integrates to 1 over its entire domain.
- Errors in integration limits or the function itself.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Always verify the two properties of a PDF.
- Use integration to find probabilities over intervals.
- If given a CDF, differentiate to find the PDF.

Probability Distribution

Definition and Core Idea: A probability distribution is a mathematical function that describes all possible values and likelihoods that a random variable can take within a given range. It can be discrete (Probability Mass Function, PMF) or continuous (Probability Density Function, PDF).

- **Important Formulas, Theorems, and Results:**

1. **For Discrete Distributions (PMF $P(X = x)$):**

- $0 \leq P(X = x) \leq 1$ for all x .
- $\sum_x P(X = x) = 1$.

2. **For Continuous Distributions (PDF $f(x)$):**

- $f(x) \geq 0$ for all x .
- $\int_{-\infty}^{\infty} f(x)dx = 1$.

3. **Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x)$.

- For discrete: $F(x) = \sum_{t \leq x} P(X = t)$.
- For continuous: $F(x) = \int_{-\infty}^x f(t)dt$.

- **Key Properties and Identities:**

- Completely characterizes a random variable's behavior.
- CDF is non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$.
- Allows calculation of probabilities, expected values, and variances.

- **Common Pitfalls or Tricky Points:**

- Confusing PMF and PDF.
- Incorrectly applying summation for continuous or integration for discrete.
- Not understanding the difference between $P(X = x)$ and $P(X \leq x)$.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Identify whether the random variable is discrete or continuous.
- Use the appropriate function (PMF/PDF) and its properties to solve problems.
- Understand the relationship between PMF/PDF and CDF.

Random Variable

Definition and Core Idea: A random variable is a variable whose value is a numerical outcome of a random phenomenon. It is a function that maps outcomes from a sample space to real numbers. Random variables can be discrete (taking countable values) or continuous (taking values in an interval).

- **Important Formulas, Theorems, and Results:**

1. **Discrete Random Variable:** Takes on a finite or countably infinite number of values (e.g., number of heads in coin flips).
2. **Continuous Random Variable:** Takes on any value within a given interval (e.g., height, temperature).
3. **Probability Distribution:** Each random variable has an associated probability distribution (PMF or PDF) that describes the probabilities of its possible values.

- **Key Properties and Identities:**

- Transforms qualitative outcomes into quantitative values.
- Allows for mathematical analysis of random phenomena.
- The sum, difference, product, or quotient of random variables is also a random variable.
- **Common Pitfalls or Tricky Points:**
 - Confusing the random variable itself with the values it can take.
 - Not correctly defining the sample space before defining the random variable.
 - Misidentifying whether a variable is discrete or continuous.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Clearly define the random variable and its possible values.
 - Determine if it's discrete or continuous to choose the correct probability distribution type.
 - Map real-world scenarios to appropriate random variable definitions.

Square Invariant

Definition and Core Idea: While "Square Invariant" is not a standard term for a probability distribution or theorem, in the context of probability and statistics, it most likely refers to properties or transformations related to the square of a random variable, particularly in the calculation of moments or variance, or in the context of distributions like the Chi-Square distribution which involves sums of squared normal variables. It implies that certain properties hold true even after squaring the variable or that a transformation involving squaring preserves some characteristic.

- **Important Formulas, Theorems, and Results:**
 1. **Second Moment:** $E[X^2] = \sum x^2 P(X = x)$ (discrete) or $\int x^2 f(x)dx$ (continuous). This is a key "square" quantity.
 2. **Variance Definition:** $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$. This formula shows how $E[X^2]$ is used to calculate variance.
 3. **Chi-Square Distribution:** Defined as the sum of squares of independent standard normal random variables, $X = \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$. This is a direct application of squaring random variables.
 4. **Properties of Variance:** $\text{Var}(aX) = a^2 \text{Var}(X)$. This shows how scaling affects the variance (a square property).
- **Key Properties and Identities:**
 - $E[X^2]$ is always non-negative.
 - Variance measures the spread of a distribution in squared units.
 - The Chi-Square distribution is fundamental for statistical inference involving variances.
- **Common Pitfalls or Tricky Points:**
 - Confusing $E[X^2]$ with $(E[X])^2$. They are distinct and generally not equal.
 - Incorrectly calculating $E[X^2]$ by not squaring the values before applying expectation.
 - Misinterpreting the degrees of freedom in Chi-Square related problems.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - When calculating variance, always compute $E[X]$ and $E[X^2]$ separately.
 - Recognize that problems involving sums of squared normal variables point towards the Chi-Square distribution.
 - Remember that variance is always non-negative.

Statistics

Definition and Core Idea: Statistics is the science of collecting, analyzing, interpreting, presenting, and organizing data. In the context of probability, statistics involves using data from a sample to make inferences about a larger population, often relying on probability distributions to quantify uncertainty.

- **Important Formulas, Theorems, and Results:**
 1. **Measures of Central Tendency:**
 - **Mean (μ or $\bar{x}$):** Average value.
 - **Median:** Middle value when data is ordered.
 - **Mode:** Most frequent value.
 2. **Measures of Dispersion:**
 - **Variance (σ^2 or s^2):** Average of the squared differences from the mean.
 - **Standard Deviation (σ or s):** Square root of variance, in original units.
 - **Range:** Max value - Min value.
 3. **Sample vs. Population:**
 - Population Mean: $\mu = E[X]$
 - Sample Mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

- Population Variance: $\sigma^2 = E[(X - \mu)^2]$
- Sample Variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ (unbiased estimator)

- Key Properties and Identities:**

- Statistics provides tools to summarize and interpret data.
- Probability distributions are models for the underlying data generation process.
- Central Limit Theorem is a bridge between probability and statistics.

- Common Pitfalls or Tricky Points:**

- Confusing population parameters (μ, σ^2) with sample statistics ($\bar{x}, s^2$).
- Using n instead of $n - 1$ in the denominator for sample variance (for unbiased estimation).
- Misinterpreting the meaning of different measures (e.g., mean vs. median for skewed data).

- Standard Problem-Solving Techniques or Shortcuts:**

- Identify whether the problem refers to a population or a sample.
- Know the definitions and formulas for common descriptive statistics.
- Understand how probability distributions are used to model observed data.

Uniform Distribution

Definition and Core Idea: The Uniform distribution is a continuous probability distribution where all values within a given interval $[a, b]$ are equally likely. It has a constant probability density over its support and zero elsewhere.

- Important Formulas, Theorems, and Results:**

- Probability Density Function (PDF):** For $X \sim U(a, b)$,

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- Cumulative Distribution Function (CDF):**

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

- Mean (Expectation):** $E[X] = \frac{a+b}{2}$

- Variance:** $\text{Var}(X) = \frac{(b-a)^2}{12}$

- Key Properties and Identities:**

- All outcomes within the interval $[a, b]$ are equally probable.
- Often used as a null hypothesis or a simple model when no other information is available.

- Common Pitfalls or Tricky Points:**

- Incorrectly identifying the interval $[a, b]$.
- Errors in calculating probabilities for sub-intervals (e.g., $P(c \leq X \leq d) = \frac{d-c}{b-a}$).
- Forgetting the constant $1/12$ in the variance formula.

- Standard Problem-Solving Techniques or Shortcuts:**

- Draw the PDF as a rectangle for visualization.
- Probabilities are simply the ratio of the length of the sub-interval to the length of the total interval.
- Memorize mean and variance formulas.

Variance

Definition and Core Idea: Variance is a measure of the spread or dispersion of a set of data points around their mean. It quantifies how much the values of a random variable deviate from its expected value, on average. A higher variance indicates greater variability.

- Important Formulas, Theorems, and Results:**

- Definition:** $\text{Var}(X) = E[(X - E[X])^2]$
- Computational Formula:** $\text{Var}(X) = E[X^2] - (E[X])^2$
- For Discrete Random Variable X :**

$$\text{Var}(X) = \sum_x (x - E[X])^2 P(X = x) = \sum_x x^2 P(X = x) - (E[X])^2$$

4. For Continuous Random Variable X :

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - E[X])^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - (E[X])^2$$

5. Properties of Variance:

- $\text{Var}(c) = 0$ for a constant c .
- $\text{Var}(aX) = a^2 \text{Var}(X)$ for a constant a .
- $\text{Var}(X + c) = \text{Var}(X)$ for a constant c .
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- For independent random variables X, Y : $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.
- For independent random variables X, Y : $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$.

6. Standard Deviation: $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

• Key Properties and Identities:

- Variance is always non-negative.
- It is expressed in squared units of the random variable.
- It measures the average squared deviation from the mean.

• Common Pitfalls or Tricky Points:

- Forgetting to square the deviations or the constant a when applying properties.
- Assuming $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ when X and Y are not independent (this requires covariance terms).
- Confusing variance with standard deviation.

• Standard Problem-Solving Techniques or Shortcuts:

- Always use the computational formula $\text{Var}(X) = E[X^2] - (E[X])^2$ as it's often easier.
- Leverage the properties of variance to simplify calculations for linear transformations or sums of independent variables.
- Calculate $E[X]$ first, then $E[X^2]$.

Quick Formula Reference

• Basic Probability

- Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Complement Rule: $P(A^c) = 1 - P(A)$

• Conditional Probability

- Definition: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Multiplication Rule: $P(A \cap B) = P(A|B)P(B)$
- Law of Total Probability: $P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$

• Independent Events

- Condition: $P(A \cap B) = P(A)P(B)$ or $P(A|B) = P(A)$

• Bayes Theorem

- Formula: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

• Expectation (Mean)

- Discrete: $E[X] = \sum_x xP(X = x)$
- Continuous: $E[X] = \int_{-\infty}^{\infty} xf(x)dx$
- Linearity: $E[aX + bY] = aE[X] + bE[Y]$

• Variance

- Definition: $\text{Var}(X) = E[(X - E[X])^2]$
- Computational: $\text{Var}(X) = E[X^2] - (E[X])^2$
- Properties: $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- For independent X, Y : $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$
- Standard Deviation: $\text{SD}(X) = \sqrt{\text{Var}(X)}$

• Bernoulli Distribution ($X \in \{0, 1\}$, $P(X = 1) = p$)

- PMF: $P(X = x) = p^x(1 - p)^{1-x}$
- Mean: $E[X] = p$
- Variance: $\text{Var}(X) = p(1 - p)$

• Binomial Distribution ($X \sim B(n, p)$)

- PMF: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
- Mean: $E[X] = np$
- Variance: $\text{Var}(X) = np(1 - p)$
- **Poisson Distribution** ($X \sim \text{Pois}(\lambda)$)
 - PMF: $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$
 - Mean: $E[X] = \lambda$
 - Variance: $\text{Var}(X) = \lambda$
- **Uniform Distribution** ($X \sim U(a, b)$)
 - PDF: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$, else 0
 - Mean: $E[X] = \frac{a+b}{2}$
 - Variance: $\text{Var}(X) = \frac{(b-a)^2}{12}$
- **Exponential Distribution** ($X \sim \text{Exp}(\lambda)$)
 - PDF: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, else 0
 - CDF: $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$
 - Mean: $E[X] = \frac{1}{\lambda}$
 - Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$
 - Memoryless: $P(X > s + t | X > s) = P(X > t)$
- **Normal Distribution** ($X \sim N(\mu, \sigma^2)$)
 - PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
 - Mean: $E[X] = \mu$
 - Variance: $\text{Var}(X) = \sigma^2$
 - Standardization: $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$
- **Chi-Square Distribution** ($X \sim \chi^2(k)$)
 - Definition: Sum of k squared independent standard normal variables.
 - Mean: $E[X] = k$
 - Variance: $\text{Var}(X) = 2k$

Important Tips for GATE

1. **Master the Fundamentals:** Ensure a strong grasp of basic probability axioms, conditional probability, and independence. Many complex problems are built upon these foundational concepts. Don't rush through them.
2. **Understand Discrete vs. Continuous:** Clearly distinguish between discrete and continuous random variables. Know when to use summation (PMF) versus integration (PDF), and remember that $P(X = x) = 0$ for continuous variables.
3. **Memorize Key Distribution Properties:** For each distribution (Bernoulli, Binomial, Poisson, Uniform, Exponential, Normal, Chi-Square), memorize their PMF/PDF, mean, and variance. This saves crucial time in the exam.
4. **Practice Conditional Probability and Bayes' Theorem:** These are frequently tested. Practice problems that require careful identification of events and application of the formulas, especially those involving multiple stages or diagnostic scenarios.
5. **Pay Attention to Keywords:** Words like "at least," "at most," "exactly," "given that," "independent," and "without replacement" significantly alter problem interpretation and solution approach. Read questions carefully.
6. **Leverage Linearity of Expectation and Variance Properties:** $E[aX + bY] = aE[X] + bE[Y]$ is powerful as it holds even for dependent variables. For variance, remember $\text{Var}(aX + b) = a^2 \text{Var}(X)$ and $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$ only for independent variables.
7. **Don't Confuse Mean and Variance Parameters:** For Exponential distribution, λ is the rate, mean is $1/\lambda$. For Normal, μ is mean, σ^2 is variance (not σ). Small errors here are common pitfalls.
8. **Practice Problem Solving:** The best way to prepare is to solve a wide variety of problems from previous GATE papers and standard textbooks. Focus on understanding the logic behind each step, not just getting the answer.



Consider Guwahati, (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and

$P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi. The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D | H_G) = 0.40$. Similarly, the next two entries are $P(M_D | H_G) = 0.48$ and $P(L_D | H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

gatecse-2018 probability bayes-theorem conditional-probability numerical-answers two-marks

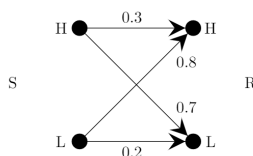
Answer key

7.1.2 Bayes Theorem: GATE CSE 2021 | Set 1 | Question: 54



A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).

In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H , is 0.7.



If the received signal is H , the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability bayes-theorem conditional-probability numerical-answers two-marks

Answer key

7.1.3 Bayes Theorem: GATE DA 2025 | Question: 21



There are three boxes containing white balls and black balls.

- Box -1 contains 2 black and 1 white balls.
- Box- 2 contains 1 black and 2 white balls.
- Box -3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box- 1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box- 3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____ (Round off to two decimal places)

gateda-2025 probability bayes-theorem numerical-answers one-mark

Answer key

7.2.1 Bayesian Network: GATE DS&AI 2024 | Question: 14



Consider five random variables U, V, W, X , and Y whose joint distribution satisfies:

$$P(U, V, W, X, Y) = P(U)P(V)P(W | U, V)P(X | W)P(Y | W)$$

Which ONE of the following statements is FALSE?

- A. Y is conditionally independent of V given W
- B. X is conditionally independent of U given W
- C. U and V are conditionally independent given W
- D. Y and X are conditionally independent given W

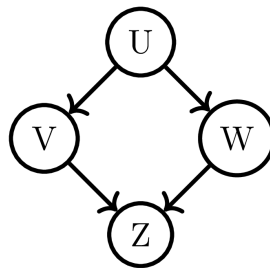
gate-ds-ai-2024 probability random-variable bayesian-network one-mark

Answer key

7.2.2 Bayesian Network: GATE DS&AI 2024 | Question: 54



Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \cdot)$	$P(V = 1 \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \cdot)$	$P(W = 1 \cdot)$
$U = 0$	1	0
$U = 1$	0	1

		$P(Z = 0 \mid \cdot)$	$P(Z = 1 \mid \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) = \underline{\hspace{2cm}}$ (rounded off to three decimal places).

gate-ds-ai-2024 probability bayesian-network numerical-answers two-marks

Answer key

7.3

Bernoulli Distribution (2)

7.3.1 Bernoulli Distribution: GATE DA 2025 | Question: 30



A random variable X is said to be distributed as $\text{Bernoulli}(\theta)$, denoted by $X \sim \text{Bernoulli}(\theta)$, if

$$P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta$$

for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim \text{Bernoulli}(\theta)$, $i = 1, 2, \dots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$, after approximation through Central Limit Theorem, is given by (Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$)

- A. $\phi(2) - \phi(-2)$ B. $\phi(1) - \phi(-1)$
 C. $\phi(3) - \phi(-3)$ D. $\phi(90) - \phi(60)$

gateda-2025 probability random-variable bernoulli-distribution two-marks

Answer key

7.3.2 Bernoulli Distribution: GATE DA 2026 | Question: 54



Let $A_{5 \times 5}$ be a matrix such that each of its elements follows $\text{Bernoulli}(p = 0.50)$ distribution independently.

The probability that the row-sum of the second row and the column-sum of the third column are both equal to 3 is _____ (Rounded off to two decimal places)

gateda-2026 probability bernoulli-distribution numerical-answers two-marks

Answer key

7.4

Binomial Distribution (6)

Practice Test: Test 1 (10Q)

7.4.1 Binomial Distribution: GATE CSE 2002 | Question: 2.16



Four fair coins are tossed simultaneously. The probability that at least one head and one tail turn up is

- A. $\frac{1}{16}$ B. $\frac{1}{8}$ C. $\frac{7}{8}$ D. $\frac{15}{16}$

gatecse-2002 probability easy binomial-distribution

Answer key

7.4.2 Binomial Distribution: GATE CSE 2005 | Question: 52



A random bit string of length n is constructed by tossing a fair coin n times and setting a bit to 0 or 1 depending on outcomes head and tail, respectively. The probability that two such randomly generated strings are not identical is:

- A. $\frac{1}{2^n}$ B. $1 - \frac{1}{n}$ C. $\frac{1}{n!}$ D. $1 - \frac{1}{2^n}$

gatecse-2005 probability binomial-distribution easy

Answer key

7.4.3 Binomial Distribution: GATE CSE 2006 | Question: 21



For each element in a set of size $2n$, an unbiased coin is tossed. The $2n$ coin tosses are independent. An element is chosen if the corresponding coin toss was a head. The probability that exactly n elements are chosen is

- A. $\frac{{}^{2n}C_n}{4^n}$ B. $\frac{{}^{2n}C_n}{2^n}$ C. $\frac{1}{{}^{2n}C_n}$ D. $\frac{1}{2}$

gatecse-2006 probability binomial-distribution normal

Answer key

7.4.4 Binomial Distribution: GATE IT 2005 | Question: 32



An unbiased coin is tossed repeatedly until the outcome of two successive tosses is the same. Assuming that the trials are independent, the expected number of tosses is

- A. 3 B. 4 C. 5 D. 6

gateit-2005 probability binomial-distribution expectation normal

Answer key

7.4.5 Binomial Distribution: GATE IT 2006 | Question: 22



When a coin is tossed, the probability of getting a Head is p , $0 < p < 1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the expected value of N is

- A. $\frac{1}{p}$ B. $\frac{1}{(1-p)}$ C. $\frac{1}{p^2}$ D. $\frac{1}{(1-p^2)}$

gateit-2006 probability binomial-distribution expectation normal

Answer key

7.4.6 Binomial Distribution: GATE IT 2007 | Question: 1



Suppose there are two coins. The first coin gives heads with probability $\frac{5}{8}$ when tossed, while the second coin gives heads with probability $\frac{1}{4}$. One of the two coins is picked up at random with equal probability and tossed. What is the probability of obtaining heads ?

- A. $\left(\frac{7}{8}\right)$ B. $\left(\frac{1}{2}\right)$ C. $\left(\frac{7}{16}\right)$ D. $\left(\frac{5}{32}\right)$

gateit-2007 probability normal binomial-distribution

Answer key

7.5

Chi Square Distribution (1)

7.5.1 Chi Square Distribution: GATE DA 2026 | Question: 43



Let $X_1, X_2, \dots, X_n$ be n independent random variables. Each of the random variables follows $\text{Normal}(\mu = 0, \sigma^2 = 1)$ distribution. Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Which of the following statements is/are correct?

- A. $\sum_{i=1}^n X_i^2$ follows Chi-square distribution with n degrees of freedom.
 B. $\sum_{i=1}^n (X_i - \bar{X})^2$ follows Chi-square distribution with $(n - 1)$ degrees of freedom.
 C. $X_1^2 + X_n^2$ follows exponential distribution with mean 2.
 D. $(\sqrt{n}\bar{X})^2$ follows Chi-square distribution with 2 degrees of freedom.

gateda-2026 random-variable normal-distribution chi-square-distribution multiple-selects two-marks

Answer key

7.6

Conditional Probability (14)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q)

7.6.1 Conditional Probability: GATE CSE 1994 | Question: 1.4, ISRO2017-2



Let A and B be any two arbitrary events, then, which one of the following is TRUE?

- A. $P(A \cap B) = P(A)P(B)$
 B. $P(A \cup B) = P(A) + P(B)$
 C. $P(A | B) = P(A \cap B)P(B)$
 D. $P(A \cup B) \leq P(A) + P(B)$

gate1994 probability conditional-probability normal isro2017

Answer key

7.6.2 Conditional Probability: GATE CSE 1994 | Question: 2.6



The probability of an event B is P_1 . The probability that events A and B occur together is P_2 while the probability that A and $\bar{B}$ occur together is P_3 . The probability of the event A in terms of P_1, P_2 and P_3 is

gate1994 probability normal conditional-probability fill-in-the-blanks

Answer key

7.6.3 Conditional Probability: GATE CSE 2003 | Question: 3



Let $P(E)$ denote the probability of the event E . Given $P(A) = 1, P(B) = \frac{1}{2}$, the values of $P(A | B)$ and $P(B | A)$ respectively are

- A. $\left(\frac{1}{4}\right), \left(\frac{1}{2}\right)$
 B. $\left(\frac{1}{2}\right), \left(\frac{1}{4}\right)$
 C. $\left(\frac{1}{2}\right), 1$
 D. $1, \left(\frac{1}{2}\right)$

gatecse-2003 probability easy conditional-probability

Answer key

7.6.4 Conditional Probability: GATE CSE 2005 | Question: 51



Box P has 2 red balls and 3 blue balls and box Q has 3 red balls and 1 blue ball. A ball is selected as follows: (i) select a box (ii) choose a ball from the selected box such that each ball in the box is equally likely

to be chosen. The probabilities of selecting boxes P and Q are $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Given that a ball selected in the above process is a red ball, the probability that it came from the box P is:

- A. $\frac{4}{19}$
 B. $\frac{5}{19}$
 C. $\frac{2}{9}$
 D. $\frac{19}{30}$

gatecse-2005 probability conditional-probability normal

Answer key

7.6.5 Conditional Probability: GATE CSE 2008 | Question: 27



Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

A. 0.24

B. 0.36

C. 0.4

D. 0.6

gatecse-2008 probability normal conditional-probability

Answer key

7.6.6 Conditional Probability: GATE CSE 2009 | Question: 21



An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same. If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3?

A. 0.453

B. 0.468

C. 0.485

D. 0.492

gatecse-2009 probability normal conditional-probability

Answer key

7.6.7 Conditional Probability: GATE CSE 2011 | Question: 3



If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

A. $\left(\frac{1}{3}\right)$ B. $\left(\frac{1}{4}\right)$ C. $\left(\frac{1}{2}\right)$ D. $\left(\frac{2}{3}\right)$

gatecse-2011 probability easy conditional-probability

Answer key

7.6.8 Conditional Probability: GATE CSE 2012 | Question: 33



Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2, or 3, the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6?

A. $\frac{10}{21}$ B. $\frac{5}{12}$ C. $\frac{2}{3}$ D. $\frac{1}{6}$

gatecse-2012 probability conditional-probability normal

Answer key

7.6.9 Conditional Probability: GATE CSE 2016 | Set 2 | Question: 05



Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____.

gatecse-2016-set2 probability conditional-probability normal numerical-answers

Answer key

7.6.10 Conditional Probability: GATE CSE 2017 | Set 2 | Question: 26



P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for this job is

A. $\left(\frac{4}{5}\right)$ B. $\left(\frac{5}{6}\right)$ C. $\left(\frac{7}{8}\right)$ D. $\left(\frac{11}{12}\right)$

gatecse-2017-set2 probability conditional-probability

Answer key

7.6.11 Conditional Probability: GATE CSE 2024 | Set 1 | Question: 53



A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____.

gatecse-2024-set1 numerical-answers probability conditional-probability two-marks

Answer key

7.6.12 Conditional Probability: GATE CSE 2026 | Set 2 | Question: 53



Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E_1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E_2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails.

The conditional probability $P(E_1 | E_2)$ is equal to _____. (rounded off to one decimal place)

gatecse-2026-set2 conditional-probability numerical-answers two-marks

Answer key

7.6.13 Conditional Probability: GATE DA 2026 | Question: 47



A clinic specializes in testing for a disease D . The result of the test can be either positive or negative.

A study revealed that if a person suffers from the disease D , the test result in that clinic comes out positive 80% of the time, and negative 20% of the time. If a person is not suffering from the disease D , the test comes out positive 10% of the time and negative 90% of the time. It is also known that among the general population, the disease D occurs in 30% of the individuals.

If a person tests positive for D in that clinic, the probability that he/she actually suffers from the disease D is _____. (Rounded off to two decimal places)

gateda-2026 probability conditional-probability numerical-answers two-marks

Answer key

7.6.14 Conditional Probability: GATE IT 2006 | Question: 1



In a certain town, the probability that it will rain in the afternoon is known to be 0.6. Moreover, meteorological data indicates that if the temperature at noon is less than or equal to $25^\circ C$, the probability that it will rain in the afternoon is 0.4. The temperature at noon is equally likely to be above $25^\circ C$, or at/below $25^\circ C$. What is the probability that it will rain in the afternoon on a day when the temperature at noon is above $25^\circ C$?

- A. 0.4 B. 0.6 C. 0.8 D. 0.9

gateit-2006 probability normal conditional-probability

Answer key

7.7 Continuous Distribution (1)

7.7.1 Continuous Distribution: GATE CSE 2016 | Set 1 | Question: 04



A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____.

gatecse-2016-set1 probability normal numerical-answers continuous-distribution

Answer key

7.8 Expectation (15)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(3Q\)](#)

7.8.1 Expectation: GATE CSE 1999 | Question: 1.1



Suppose that the expectation of a random variable X is 5. Which of the following statements is true?

- A. There is a sample point at which X has the value 5.
- B. There is a sample point at which X has value greater than 5.
- C. There is a sample point at which X has a value greater than equal to 5.
- D. None of the above.

gate1999 probability expectation easy

Answer key

7.8.2 Expectation: GATE CSE 2004 | Question: 74



An examination paper has 150 multiple choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 marks. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

- A. 0
- B. 2550
- C. 7525
- D. 9375

gatecse-2004 probability expectation normal

Answer key

7.8.3 Expectation: GATE CSE 2006 | Question: 18



We are given a set $X = \{X_1, \dots, X_n\}$ where $X_i = 2^i$. A sample $S \subseteq X$ is drawn by selecting each X_i independently with probability $P_i = \frac{1}{2}$. The expected value of the smallest number in sample S is:

- A. $(\frac{1}{n})$
- B. 2
- C. $\sqrt{n}$
- D. n

gatecse-2006 probability expectation normal

Answer key

7.8.4 Expectation: GATE CSE 2011 | Question: 18



If the difference between the expectation of the square of a random variable ($E[X^2]$) and the square of the expectation of the random variable ($E[X]^2$) is denoted by R , then

- A. $R = 0$
- B. $R < 0$
- C. $R \geq 0$
- D. $R > 0$

gatecse-2011 probability random-variable expectation normal

Answer key

7.8.5 Expectation: GATE CSE 2013 | Question: 24



Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is $\frac{1}{2}$. What is the expected number of unordered cycles of length three?

- A. $\frac{1}{8}$
- B. 1
- C. 7
- D. 8

gatecse-2013 probability expectation normal

Answer key

7.8.6 Expectation: GATE CSE 2014 | Set 2 | Question: 2



Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is _____. (The answer should be rounded to one decimal place.)

gatecse-2014-set2 probability expectation numerical-answers easy

7.8.10 Expectation: GATE CSE 2026 | Set 1 | Question: 48



Let X be a random variable which takes values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Further, $\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 5) = \Pr(X = 7) = \frac{1}{6}$ and $\Pr(X = 3) = \Pr(X = 4) = \Pr(X = 6) = \Pr(X = 8) = \frac{1}{12}$.

The expected value of X , denoted by $E[X]$, is equal to _____. (rounded off to two decimal places)

gatecse-2026-set1 numerical-answers two-marks probability expectation

Answer key

7.8.11 Expectation: GATE DA 2025 | Question: 1



Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X | Y]$. Then $E[E[X | Y]]$ equals

- A. $E[X | Y]$ B. $\frac{E[X]}{E[Y]}$ C. $E[X]$ D. $E[Y]$

gateda-2025 probability random-variable expectation one-mark

Answer key

7.8.12 Expectation: GATE DA 2025 | Question: 10



Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4$$

where $E[X]$ denotes the expectation of random variable X . The values of a, b are:

- A. $a = -2, b = 1$ B. $a = 2, b = -1$ C. $a = -2, b = -1$ D. $a = 1, b = 1$

gateda-2025 probability expectation random-variable one-mark

Answer key

7.8.13 Expectation: GATE DA 2025 | Question: 51



A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment.

The expectation of S_{100} denoted by $E[S_{100}] = \underline{\hspace{2cm}}$. (Round off to one decimal place)

gateda-2025 probability expectation numerical-answers two-marks

Answer key

7.8.14 Expectation: GATE DS&AI 2024 | Question: 26



A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently.

What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?

- A. 2 B. 4 C. 6 D. 8

gate-ds-ai-2024 probability expectation two-marks

Answer key

7.8.15 Expectation: GATE DS&AI 2024 | Question: 49



Consider a joint probability density function of two random variables X and Y

$$f_{X,Y}(x,y) = \begin{cases} 2xy, & 0 < x < 2, \quad 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

Then, $E[Y \mid X = 1.5]$ is _____

gate-ds-ai-2024 probability expectation numerical-answers two-marks

Answer key

7.9

Exponential Distribution (6)

 **Practice Test:** Test 1 (7Q)

7.9.1 Exponential Distribution: GATE CSE 2021 | Set 1 | Question: 18



The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is _____.

gatecse-2021-set1 probability random-variable numerical-answers one-mark exponential-distribution

Answer key

7.9.2 Exponential Distribution: GATE DA 2025 | Question: 11



It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X . What is the value of λ ? (ln denotes natural logarithm)

- A. $\ln 2$ B. $\ln 4$ C. $\ln 3$ D. $\ln 0.25$

gateda-2025 probability exponential-distribution expectation one-mark

Answer key

7.9.3 Exponential Distribution: GATE DA 2025 | Question: 31



For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k + 1,$$

where k is an integer. Let $Y = \lfloor X \rfloor$, where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where $\ln$ denotes natural logarithm. For any positive integer ℓ , one can write the probability of the event $Y = \ell$ as follows

$$P(Y = \ell) = q^\ell (1 - q)$$

The value of q is

- A. 0.1 B. 0.01 C. 0.5 D. 0.434

gateda-2025 probability random-variable exponential-distribution two-marks

Answer key

7.9.4 Exponential Distribution: GATE DA 2026 | Question: 24



Let X be an exponentially distributed random variable with mean $\lambda (> 0)$. If $P(X > 5) = 0.35$, then the conditional probability $P(X > 10 \mid X > 5)$ is _____. (Rounded off to two decimal places)

gateda-2026 conditional-probability random-variable exponential-distribution numerical-answers one-mark

Answer key

7.9.5 Exponential Distribution: GATE DS&AI 2024 | Question: 47



Let X be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ indicate the expectation and variance of X , respectively, the value of λ is _____ (rounded off to one decimal place).

gate-ds-ai-2024 numerical-answers probability random-variable exponential-distribution two-marks

Answer key

7.9.6 Exponential Distribution: GATE IT 2004 | Question: 33



Let X and Y be two exponentially distributed and independent random variables with mean α and β , respectively. If $Z = \min(X, Y)$, then the mean of Z is given by

- A. $\left(\frac{1}{\alpha + \beta}\right)$ B. $\min(\alpha, \beta)$
 C. $\left(\frac{\alpha\beta}{\alpha + \beta}\right)$ D. $\alpha + \beta$

gateit-2004 probability exponential-distribution random-variable normal

Answer key

7.10

Independent Events (6)

Practice Test: Test 1 (13Q)

7.10.1 Independent Events: GATE CSE 1994 | Question: 2.8



Let A , B , and C be independent events which occur with probabilities 0.8, 0.5, and 0.3 respectively. The probability of occurrence of at least one of the event is _____

gate1994 probability normal numerical-answers independent-events

Answer key

7.10.2 Independent Events: GATE CSE 1999 | Question: 2.1



Consider two events E_1 and E_2 such that probability of E_1 , $P_r[E_1] = \frac{1}{2}$, probability of E_2 , $P_r[E_2] = \frac{1}{3}$, and probability of E_1 and E_2 , $P_r[E_1 \text{ and } E_2] = \frac{1}{5}$. Which of the following statements is/are true?

- A. $P_r[E_1 \text{ or } E_2]$ is $\frac{2}{3}$ B. Events E_1 and E_2 are independent
 C. Events E_1 and E_2 are not independent D. $P_r[E_1 | E_2] = \frac{4}{5}$

gate1999 probability normal independent-events

Answer key

7.10.3 Independent Events: GATE CSE 2000 | Question: 2.2



E_1 and E_2 are events in a probability space satisfying the following constraints:

- $Pr(E_1) = Pr(E_2)$
- $Pr(E_1 \cup E_2) = 1$
- E_1 and E_2 are independent

The value of $Pr(E_1)$, the probability of the event E_1 , is

- A. 0 B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. 1

gatecse-2000 probability easy independent-events

Answer key

7.10.4 Independent Events: GATE CSE 2023 | Question: 43



Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

- A. A and B are independent.
- B. A and C are independent.
- C. B and C are independent.
- D. $\text{Prob}(B | C) = \text{Prob}(B)$

gategse-2023 probability independent-events multiple-selects two-marks

Answer key

7.10.5 Independent Events: GATE DA 2025 | Question: 44



Consider a coin-toss experiment where the probability of head showing up is p . In the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

where n is the total number of independent coin tosses.

Which of the following statements is/are correct?

- A. $E[\hat{p}] = p$
- B. $E[\hat{p}] = \frac{p}{n}$
- C. As n increases, variance of $\hat{p}$ decreases
- D. Variance of $\hat{p}$ does not depend on n

gateda-2025 probability independent-events multiple-selects two-marks

Answer key

7.10.6 Independent Events: GATE DS&AI 2024 | Question: 2



Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails.

What is the probability of the event $T \cap S$?

- A. 0
- B. 0.5
- C. 0.25
- D. 1

gate-ds-ai-2024 probability independent-events one-mark

Answer key

7.11 Normal Distribution (4)

Practice Test: Test 1 (7Q)

7.11.1 Normal Distribution: GATE CSE 2008 | Question: 29



Let X be a random variable following normal distribution with mean $+1$ and variance 4 . Let Y be another normal variable with mean -1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

- A. 3
- B. 2
- C. $\sqrt{2}$
- D. 1

gategse-2008 random-variable normal-distribution probability normal

Answer key

7.11.2 Normal Distribution: GATE CSE 2017 Set-1 | Question: 37



Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

Answer key

7.11.3 Normal Distribution: GATE CSE 2017 | Set 1 | Question: 19



Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

gatecse-2017-set1 probability numerical-answers normal-distribution

Answer key

7.11.4 Normal Distribution: GATE DA 2026 | Question: 18



Suppose a random variable Z follows $\text{Normal}(\mu = 0, \sigma^2 = 1)$ distribution with probability density function $g(z)$ and cumulative distribution function $G(z)$. Another random variable Y follows t_1 distribution with probability density function $h(y)$ and cumulative distribution function $H(y)$. Let c be the positive real number for which $g(c) = h(c)$.

Which of the following statements is/are correct?

- A. $G(0) = H(0)$ B. $G(c) < H(c)$ C. $G(-c) < H(-c)$ D. $g(0) = h(0)$

gateda-2026 probability random-variable normal-distribution multiple-selects one-mark

Answer key

7.12

Poisson Distribution (5)

Practice Test: Test 1 (5Q)

7.12.1 Poisson Distribution: GATE CSE 1989 | Question: 4-viii



$P_n(t)$ is the probability of n events occurring during a time interval t . How will you express $P_0(t+h)$ in terms of $P_0(h)$, if $P_0(t)$ has stationary independent increments? (Note: $P_t(t)$ is the probability density function).

gate1989 descriptive probability poisson-distribution

Answer key

7.12.2 Poisson Distribution: GATE CSE 2013 | Question: 2



Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

- A. $\frac{8}{(2e^3)}$ B. $\frac{9}{(2e^3)}$ C. $\frac{17}{(2e^3)}$ D. $\frac{26}{(2e^3)}$

gatecse-2013 probability poisson-distribution normal

Answer key

7.12.3 Poisson Distribution: GATE CSE 2017 | Set 2 | Question: 48



If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(x+2)^2]$ equals _____.

gatecse-2017-set2 expectation poisson-distribution numerical-answers probability

Answer key

7.12.4 Poisson Distribution: GATE DA 2026 | Question: 35



Let

$$L = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{e^{-n} n^k}{k!}$$

Which of the following is the value of L ?

- A. 0.5 B. 1.0 C. 0 D. e^{-1}

gateda-2026 probability poisson-distribution two-marks

Answer key 

7.12.5 Poisson Distribution: GATE IT 2007 | Question: 57



In a multi-user operating system on an average, 20 requests are made to use a particular resource per hour. The arrival of requests follows a Poisson distribution. The probability that either one, three or five requests are made in 45 minutes is given by :

- A. $6.9 \times 10^6 \times e^{-20}$ B. $1.02 \times 10^6 \times e^{-20}$
C. $6.9 \times 10^3 \times e^{-20}$ D. $1.02 \times 10^3 \times e^{-20}$

gateit-2007 probability poisson-distribution normal

Answer key 

7.13 Probability (31)

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(15Q\)](#)

7.13.1 Probability: GATE CSE 1995 | Question: 1.18



The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is:

- A. $\frac{16}{25}$ B. $\left(\frac{9}{10}\right)^3$ C. $\frac{27}{75}$ D. $\frac{18}{25}$

gate1995 probability normal

Answer key 

7.13.2 Probability: GATE CSE 1995 | Question: 2.14



A bag contains 10 white balls and 15 black balls. Two balls are drawn in succession. The probability that one of them is black and the other is white is:

- A. $\frac{2}{3}$ B. $\frac{4}{5}$ C. $\frac{1}{2}$ D. $\frac{1}{3}$

gate1995 probability normal

Answer key 

7.13.3 Probability: GATE CSE 1996 | Question: 1.5



Two dice are thrown simultaneously. The probability that at least one of them will have 6 facing up is

- A. $\frac{1}{36}$ B. $\frac{1}{3}$ C. $\frac{25}{36}$ D. $\frac{11}{36}$

gate1996 probability easy

Answer key 

7.13.4 Probability: GATE CSE 1996 | Question: 2.7



The probability that top and bottom cards of a randomly shuffled deck are both aces is

- A. $\frac{4}{52} \times \frac{4}{52}$ B. $\frac{4}{52} \times \frac{3}{52}$
C. $\frac{4}{52} \times \frac{3}{51}$ D. $\frac{4}{52} \times \frac{4}{51}$

Answer key

7.13.5 Probability: GATE CSE 1997 | Question: 1.1



The probability that it will rain today is 0.5. The probability that it will rain tomorrow is 0.6. The probability that it will rain either today or tomorrow is 0.7. What is the probability that it will rain today and tomorrow?

- A. 0.3 B. 0.25 C. 0.35 D. 0.4

Answer key

7.13.6 Probability: GATE CSE 1998 | Question: 1.1



A die is rolled three times. The probability that exactly one odd number turns up among the three outcomes is

- A. $\frac{1}{6}$ B. $\frac{3}{8}$ C. $\frac{1}{8}$ D. $\frac{1}{2}$

Answer key

7.13.7 Probability: GATE CSE 2001 | Question: 2.4



Seven (distinct) car accidents occurred in a week. What is the probability that they all occurred on the same day?

- A. $\frac{1}{7^7}$ B. $\frac{1}{7^6}$ C. $\frac{1}{2^7}$ D. $\frac{7}{2^7}$

Answer key

7.13.8 Probability: GATE CSE 2004 | Question: 25



If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

- A. $\frac{3}{8}$ B. $\frac{1}{2}$ C. $\frac{5}{8}$ D. $\frac{3}{4}$

Answer key

7.13.9 Probability: GATE CSE 2010 | Question: 26



Consider a company that assembles computers. The probability of a faulty assembly of any computer is p . The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q . What is the probability of a computer being declared faulty?

- A. $pq + (1-p)(1-q)$ B. $(1-q)p$ C. $(1-p)q$ D. pq

Answer key

7.13.10 Probability: GATE CSE 2010 | Question: 27



What is the probability that divisor of 10^{99} is a multiple of 10^{96} ?

- A. $\left(\frac{1}{625}\right)$ B. $\left(\frac{4}{625}\right)$ C. $\left(\frac{12}{625}\right)$ D. $\left(\frac{16}{625}\right)$

Answer key

7.13.11 Probability: GATE CSE 2011 | Question: 34



A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

A. $\left(\frac{1}{5}\right)$

B. $\left(\frac{4}{25}\right)$

C. $\left(\frac{1}{4}\right)$

D. $\left(\frac{2}{5}\right)$

gatecse-2011 probability normal

Answer key

7.13.12 Probability: GATE CSE 2014 | Set 1 | Question: 48



Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $\frac{X}{1296}$. The value of X is _____

gatecse-2014-set1 probability numerical-answers normal

Answer key

7.13.13 Probability: GATE CSE 2014 | Set 2 | Question: 1



The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p . Then $100p =$ _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key

7.13.14 Probability: GATE CSE 2014 | Set 2 | Question: 48



The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is _____.

gatecse-2014-set2 probability numerical-answers normal

Answer key

7.13.15 Probability: GATE CSE 2014 | Set 3 | Question: 48



Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)P(B)$ is _____.

gatecse-2014-set3 probability numerical-answers normal

Answer key

7.13.16 Probability: GATE CSE 2016 | Set 1 | Question: 29



Consider the following experiment.

Step 1. Flip a fair coin twice.

Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places)

gatecse-2016-set1 probability normal numerical-answers

Answer key

7.13.17 Probability: GATE CSE 2018 | Question: 15



Two people, P and Q , decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q . Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____

gatecse-2018 probability normal numerical-answers one-mark

Answer key

7.13.18 Probability: GATE CSE 2021 | Set 2 | Question: 33



A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

- A. $\frac{r}{r+b}$
- B. $\frac{r}{r+b+3}$
- C. $\frac{r+3}{r+b+3}$
- D. $\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

gatecse-2021-set2 probability normal two-marks

Answer key

7.13.19 Probability: GATE CSE 2024 | Set 1 | Question: 17



Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

- A. The two events A and B are independent
- B. $P(A \cup B) = 0.7$
- C. $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- D. $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B , respectively

gatecse-2024-set1 multiple-selects probability one-mark

Answer key

7.13.20 Probability: GATE CSE 2024 | Set 2 | Question: 8



When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (*i. e.*, 1, 2, 3, 4, 5, and 6) is

- A. $\frac{1}{324}$
- B. $\frac{5}{324}$
- C. $\frac{7}{324}$
- D. $\frac{11}{324}$

gatecse-2024-set2 probability one-mark

Answer key

7.13.21 Probability: GATE CSE 2025 | Set 1 | Question: 22



A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

gatecse2025-set1 probability numerical-answers one-mark

Answer key

7.13.22 Probability: GATE CSE 2026 | Set 1 | Question: 1



An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is

- A. $1/4$ B. $1/3$ C. $1/2$ D. $2/3$

gatecse-2026-set1 probability one-mark

Answer key

7.13.23 Probability: GATE DA 2025 | Question: 35



A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

- A. 0 B. 1 C. $1 - \left(\frac{5}{6}\right)^{100}$ D. $\left(\frac{5}{6}\right)^{100}$

gateda-2025 probability two-marks

Answer key

7.13.24 Probability: GATE DA 2026 | Question: 10



Suppose that a computer program provides a non-negative and integer-valued random solution to the equation $n_1 + n_2 + n_3 + n_4 = 20$.

Which of the following is the probability that all of n_1, n_2, n_3, n_4 in the provided solution are positive?

- A. $\binom{19}{3} / \binom{23}{3}$ B. $\binom{20}{4} / \binom{24}{4}$
C. $\binom{20}{3} / \binom{23}{3}$ D. $\binom{19}{4} / \binom{24}{4}$

gateda-2026 probability one-mark

Answer key

7.13.25 Probability: GATE DA 2026 | Question: 9



Let M be a randomly chosen non-empty subset of $S = \{1, 2, 3, \dots, 2026\}$.

Which of the following is the probability that the product of all the elements of M is even?

- A. $2^{1013} (2^{1013} - 1) / 2^{2026}$ B. $2^{1013} / 2^{2026}$
C. $2^{1013} (2^{1013} - 1) / (2^{2026} - 1)$ D. $1 / (2^{2026} - 1)$

gateda-2026 probability one-mark

Answer key

7.13.26 Probability: GATE DS&AI 2024 | Question: 48



Consider two events T and S . Let $\bar{T}$ denote the complement of the event T . The probability associated with different events are given as follows:

$$P(\bar{T}) = 0.6, \quad P(S | T) = 0.3, \quad P(S | \bar{T}) = 0.6$$

Then, $P(T | S)$ is _____ (rounded off to two decimal places).

Answer key

7.13.27 Probability: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 7



A fair coin is flipped twice and it is known that at least one tail is observed. The probability of getting two tails is

- A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\frac{1}{4}$

Answer key

7.13.28 Probability: GATE IT 2004 | Question: 1



In a population of N families, 50% of the families have three children, 30% of the families have two children and the remaining families have one child. What is the probability that a randomly picked child belongs to a family with two children?

- A. $\left(\frac{3}{23}\right)$ B. $\left(\frac{6}{23}\right)$ C. $\left(\frac{3}{10}\right)$ D. $\left(\frac{3}{5}\right)$

Answer key

7.13.29 Probability: GATE IT 2005 | Question: 1



A bag contains 10 blue marbles, 20 green marbles and 30 red marbles. A marble is drawn from the bag, its colour recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour is

- A. $\left(\frac{1}{36}\right)$ B. $\left(\frac{1}{6}\right)$ C. $\left(\frac{1}{4}\right)$ D. $\left(\frac{1}{3}\right)$

Answer key

7.13.30 Probability: GATE IT 2008 | Question: 2



A sample space has two events A and B such that probabilities $P(A \cap B) = \frac{1}{2}$, $P(A') = \frac{1}{3}$, $P(B') = \frac{1}{3}$. What is $P(A \cup B)$?

- A. $\left(\frac{11}{12}\right)$ B. $\left(\frac{10}{12}\right)$ C. $\left(\frac{9}{12}\right)$ D. $\left(\frac{8}{12}\right)$

Answer key

7.13.31 Probability: GATE IT 2008 | Question: 23



What is the probability that in a randomly chosen group of r people at least three people have the same birthday?

- A. $1 - \frac{365 - 364 \dots (365 - r + 1)}{365^r}$
 B. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} + {}^r C_1 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r+2}}$
 C. $1 - \frac{365 \cdot 364 \dots (365 - r + 1)}{365^r} - {}^r C_2 \cdot 365 \cdot \frac{364 \cdot 363 \dots (364 - (r - 2) + 1)}{364^{r-2}}$
 D. $\frac{365 \cdot 364 \dots (365 - r + 1)}{365^r}$

Answer key

7.14

Probability Density Function (1)

7.14.1 Probability Density Function: GATE CSE 2003 | Question: 60, ISRO2007-45



A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

- A. $f_1(t) + f_2(t)$ B. $\int_0^t f_1(x)f_2(x)dx$
 C. $\int_0^t f_1(x)f_2(t-x)dx$ D. $\max\{f_1(t), f_2(t)\}$

gatecse-2003 probability normal isro2007 probability-density-function

Answer key

7.15

Probability Distribution (1)

7.15.1 Probability Distribution: GATE CSE 2025 | Set 1 | Question: 48



Consider a probability distribution given by the density function $P(x)$.

$$P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$

The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (rounded off to three decimal places)

gatecse2025-set1 probability probability-distribution numerical-answers two-marks

Answer key

7.16

Random Variable (10)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(11Q\)](#)

7.16.1 Random Variable: GATE CSE 2005 | Question: 12, ISRO2009-64



Let $f(x)$ be the continuous probability density function of a random variable x , the probability that $a < x \leq b$ is :

- A. $f(b-a)$ B. $f(b) - f(a)$
 C. $\int_a^b f(x)dx$ D. $\int_a^b xf(x)dx$

gatecse-2005 probability random-variable easy isro2009

Answer key

7.16.2 Random Variable: GATE CSE 2011 | Question: 33



Consider a finite sequence of random values $X = [x_1, x_2, \dots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X . Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence.

Which one of the following statements is **INCORRECT**?

- A. Index position of mode of X in X is the same as the index position of mode of Y in Y
 B. Index position of median of X in X is the same as the index position of median of Y in Y
 C. $\mu_y = a\mu_x + b$
 D. $\sigma_y = a\sigma_x + b$

Answer key

7.16.3 Random Variable: GATE CSE 2012 | Question: 21



Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

- A. 0 and 0.5 B. 0 and 1 C. 0.5 and 1 D. 0.25 and 0.75

Answer key

7.16.4 Random Variable: GATE CSE 2015 | Set 3 | Question: 37



Suppose X_i for $i = 1, 2, 3$ are independent and identically distributed random variables whose probability mass functions are $Pr[X_i = 0] = Pr[X_i = 1] = \frac{1}{2}$ for $i = 1, 2, 3$. Define another random variable $Y = X_1 X_2 \oplus X_3$, where $\oplus$ denotes XOR. Then $Pr[Y = 0 | X_3 = 0] = \underline{\hspace{2cm}}$.

Answer key

7.16.5 Random Variable: GATE CSE 2026 | Set 2 | Question: 4



The probability density function $f(x)$ of a random variable X which takes real values is

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), \quad x \in (-\infty, +\infty)$$

Which one of the following statements is correct about the random variable X ?

- A. X is an exponential random variable B. X is a normal random variable
C. X is a Poisson random variable D. X is a uniform random variable

Answer key

7.16.6 Random Variable: GATE DA 2025 | Question: 29



Consider the cumulative distribution function (CDF) of a random variable X :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases}$$

The value of $P(X^2 \leq 0.25)$ is

- A. 0.625 B. 0.25 C. 0.5 D. 0.5625

Answer key

7.16.7 Random Variable: GATE DA 2025 | Question: 9



Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

If the median of X is 3, then what is the value of t ?

- A. 2 B. 1 C. -1 D. 0

gateda-2025 probability random-variable one-mark

Answer key 

7.16.8 Random Variable: GATE DA 2026 | Question: 44



Let X be a discrete valued random variable with cumulative distribution function $F(x)$.

Which of the following statements is/are correct?

- A. $F(x)$ is always a positive function. B. $F(x)$ is a non-decreasing function.
C. $F(x)$ has jump discontinuity. D. $F(x)$ is a left continuous function.

gateda-2026 random-variable probability multiple-selects two-marks

Answer key 

7.16.9 Random Variable: GATE DS&AI 2024 | Question: 1



Consider the following statements:

- i. The mean and variance of a Poisson random variable are equal.
ii. For a standard normal random variable, the mean is zero and the variance is one.

Which **ONE** of the following options is **correct**?

- A. Both (i) and (ii) are true B. (i) is true and (ii) is false
C. (ii) is true and (i) is false D. Both (i) and (ii) are false

gate-ds-ai-2024 probability random-variable one-mark

Answer key 

7.16.10 Random Variable: GATE DS&AI 2024 | Question: 55



Two fair coins are tossed independently. X is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. Y is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise.

The value of the covariance of X and Y is _____ (rounded off to three decimal places).

gate-ds-ai-2024 probability random-variable numerical-answers two-marks

Answer key 

7.17

Square Invariant (1)

7.17.1 Square Invariant: GATE CSE 2025 | Set 2 | Question: 54



A quadratic polynomial $(x - \alpha)(x - \beta)$ over complex numbers is said to be square invariant if $(x - \alpha)(x - \beta) = (x - \alpha^2)(x - \beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random.

The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)

gatecse2025-set2 probability square-invariant numerical-answers two-marks

Answer key 

 **Practice Test:** Test 1 (12Q)

7.18.1 Statistics: GATE CSE 2021 | Set 2 | Question: 22



For a given biased coin, the probability that the outcome of a toss is a head is 0.4 . This coin is tossed $1,000$ times. Let X denote the random variable whose value is the number of times that head appeared in these $1,000$ tosses. The standard deviation of X (rounded to 2 decimal place) is _____

gatecse-2021-set2 numerical-answers probability random-variable one-mark statistics

Answer key 

7.18.2 Statistics: GATE CSE 2024 | Set 2 | Question: 34



Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

- A. $\bar{z} = \bar{x}\bar{y}$ B. $\bar{z} \leq \bar{x}\bar{y}$
C. $\bar{z} \geq \bar{x}\bar{y}$ D. $\bar{z} \leq \bar{x}$

gatecse-2024-set2 probability random-variable statistics two-marks

Answer key 

7.18.3 Statistics: GATE DA 2026 | Question: 52



For a given data set $\{x_1, x_2, \dots, x_n\}$, where $n = 100$, it is known that

$$\frac{1}{2000} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = 99$$

Let us denote $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

The value of $\frac{1}{99} \sum_{i=1}^n (x_i - \bar{x})^2$ is _____. (Answer in **integer**)

gateda-2026 numerical-answers two-marks statistics

Answer key 

 **Practice Tests:** Test 1 (15Q) Test 2 (3Q)

7.19.1 Uniform Distribution: GATE CSE 1998 | Question: 3a



Two friends agree to meet at a park with the following conditions. Each will reach the park between 4:00 pm and 5:00 pm and will see if the other has already arrived. If not, they will wait for 10 minutes or the end of the hour whichever is earlier and leave. What is the probability that the two will not meet?

gate1998 probability normal numerical-answers uniform-distribution

Answer key 

7.19.2 Uniform Distribution: GATE CSE 2004 | Question: 78



Two n bit binary strings, S_1 and S_2 are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

- A. $\frac{nC_d}{2^n}$ B. $\frac{nC_d}{2^d}$ C. $\frac{d}{2^n}$ D. $\frac{1}{2^d}$

gatecse-2004 probability normal uniform-distribution

Answer key 

7.19.3 Uniform Distribution: GATE CSE 2004 | Question: 80



A point is randomly selected with uniform probability in the $X - Y$ plane within the rectangle with corners at $(0, 0)$, $(1, 0)$, $(1, 2)$ and $(0, 2)$. If p is the length of the position vector of the point, the expected value of p^2 is

- A. $\left(\frac{2}{3}\right)$ B. 1 C. $\left(\frac{4}{3}\right)$ D. $\left(\frac{5}{3}\right)$

gatecse-2004 probability uniform-distribution expectation normal

Answer key

7.19.4 Uniform Distribution: GATE CSE 2007 | Question: 24



Suppose we uniformly and randomly select a permutation from the $20!$ permutations of $1, 2, 3, \dots, 20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

- A. $\left(\frac{1}{2}\right)$ B. $\left(\frac{1}{10}\right)$ C. $\left(\frac{9!}{20!}\right)$ D. None of these

gatecse-2007 probability easy uniform-distribution

Answer key

7.19.5 Uniform Distribution: GATE CSE 2014 | Set 1 | Question: 2



Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is _____.

gatecse-2014-set1 probability uniform-distribution expectation numerical-answers normal

Answer key

7.19.6 Uniform Distribution: GATE CSE 2019 | Question: 47



Suppose Y is distributed uniformly in the open interval $(1, 6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____

gatecse-2019 numerical-answers engineering-mathematics probability uniform-distribution two-marks

Answer key

7.19.7 Uniform Distribution: GATE CSE 2020 | Question: 45



For $n > 2$, let $a \in \{0, 1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0, 1\}^n$. Then, the probability that $\sum_{i=1}^n a_i x_i$ is an odd number is _____

gatecse-2020 numerical-answers probability uniform-distribution two-marks

Answer key

7.19.8 Uniform Distribution: GATE CSE 2024 | Set 1 | Question: 4



Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

- A. The events X and Y are mutually exclusive
B. The events X and Y are independent
C. Either event X or Y must occur
D. Event X is more likely than event Y

gatecse-2024-set1 probability uniform-distribution one-mark

Answer key

7.19.9 Uniform Distribution: GATE CSE 2025 | Set 2 | Question: 55



The unit interval $(0, 1)$ is divided at a point chosen uniformly distributed over $(0, 1)$ in $\mathbb{R}$ into two disjoint subintervals.

The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

gatecse2025-set2 probability uniform-distribution numerical-answers two-marks

Answer key

7.19.10 Uniform Distribution: GATE DA 2026 | Question: 53



Let X be a random variable that follows Uniform $(-1, 1)$ distribution. The conditional distribution of the random variable Y given $X = x$ is the Uniform $(x^2 - 0.1, x^2 + 0.1)$ distribution.

The value of correlation (X, Y) is _____. (Answer in integer)

gateda-2026 probability random-variable uniform-distribution numerical-answers two-marks

Answer key

7.19.11 Uniform Distribution: GATE DS&AI 2024 | Question: 46



Let X be a random variable uniformly distributed in the interval $[1, 3]$ and Y be a random variable uniformly distributed in the interval $[2, 4]$. If X and Y are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).

gate-ds-ai-2024 numerical-answers random-variable probability uniform-distribution two-marks

Answer key

7.20 Variance (2)

Practice Test: Test 1 (5Q)

7.20.1 Variance: GATE DA 2025 | Question: 26



Let $Y = Z^2$, $Z = \frac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2 . The variance of Y is

- A. 1 B. 2 C. 3 D. 4

gateda-2025 probability random-variable variance two-marks

Answer key

7.20.2 Variance: GATE DA 2026 | Question: 34



Let X and Y be two independent random variables. X follows Bernoulli($p = 0.3$) distribution and Y follows Normal($\mu = 0, \sigma^2 = 100$) distribution.

Which of the following options is the variance of $(2X - 1)Y$?

- A. 100 B. 90 C. 49 D. 21

gateda-2026 probability random-variable normal-distribution variance two-marks

Answer key

Answer Keys

7.1.1	0.60 : 0.61	7.1.2	0.04 : 0.04	7.1.3	0.25:0.25	7.2.1	C	7.2.2	0.125
7.3.1	A	7.3.2	0.08:0.12	7.4.1	C	7.4.2	D	7.4.3	A
7.4.4	A	7.4.5	A	7.4.6	C	7.5.1	A;B;C	7.6.1	D
7.6.2	N/A	7.6.3	D	7.6.4	A	7.6.5	C	7.6.6	B
7.6.7	A	7.6.8	B	7.6.9	0.55	7.6.10	A	7.6.11	0.375

7.6.12	0.5:0.5
7.8.2	D
7.8.7	B
7.8.12	A
7.9.2	A
7.10.1	0.93
7.10.6	A
7.12.1	N/A
7.13.1	D
7.13.6	B
7.13.11	A
7.13.16	0.33 : 0.34
7.13.21	0.49:0.51
7.13.26	0.25
7.13.31	X
7.16.3	C
7.16.8	B;C
7.18.2	D
7.19.4	B
7.19.9	0.70:0.80

7.6.13	0.76:0.78
7.8.3	D
7.8.8	D
7.8.13	66.6:66.7
7.9.3	A
7.10.2	C
7.11.1	A
7.12.2	C
7.13.2	C
7.13.7	B
7.13.12	10
7.13.17	0.0230 : 0.0232
7.13.22	B
7.13.27	B
7.14.1	C
7.16.4	0.75
7.16.9	A
7.18.3	10:10
7.19.5	0.24 : 0.27
7.19.10	0:0

7.6.14	C
7.8.4	C
7.8.9	D
7.8.14	C
7.9.4	0.34:0.36
7.10.3	D
7.11.2	TBA
7.12.3	54
7.13.3	D
7.13.8	A
7.13.13	11.85 : 11.95
7.13.18	A
7.13.23	C
7.13.28	B
7.15.1	0.300:0.302
7.16.5	B
7.16.10	0.062:0.063
7.19.1	0.69:0.70
7.19.6	0.8
7.19.11	0.125

7.7.1	0.5
7.8.5	C
7.8.10	4.24 : 4.26
7.8.15	X
7.9.5	0.2
7.10.4	C;D
7.11.3	0
7.12.4	A
7.13.4	C
7.13.9	A
7.13.14	0.259 : 0.261
7.13.19	B;C
7.13.24	A
7.13.29	B
7.16.1	C
7.16.6	C
7.17.1	0.5:0.5
7.19.2	A
7.19.7	0.5
7.20.1	B

7.8.1	C
7.8.6	3.8 : 3.9
7.8.11	C
7.9.1	0.35 : 0.39
7.9.6	C
7.10.5	A;C
7.11.4	A;C
7.12.5	B
7.13.5	D
7.13.10	A
7.13.15	0.25
7.13.20	B
7.13.25	C
7.13.30	B
7.16.2	D
7.16.7	A
7.18.1	15.00 : 16.00
7.19.3	D
7.19.8	B
7.20.2	A